

that the browser renders the page in standards mode. The DOCTYPE has no other purpose and is therefore optional for XML.

In HTML5, the DOCTYPE for web pages has been greatly simplified. Compare, for example, the following HTML4 DOCTYPEs:

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//
EN"
    "http://www.w3.org/TR/html4/loose.dtd">
```

Impossible to remember, right? HTML5 neatly solves this problem as follows:

- `<!DOCTYPE html>`

Yes, that's it! Just two words "DOCTYPE" and "html". All this is possible because HTML 5 is no longer part of SGML, but is instead a markup language all by itself.

Character declaration

Like the new DOCTYPE, the character set declaration has also been abbreviated. It used to be:

- `<meta http-equiv="Content-Type" content="text/html; charset=utf-8">`

Now, it uses UTF-8 and you define it with just one meta tag:

- `<meta charset="utf-8">`

MathML and SVG

The HTML syntax of HTML5 allows for MathML and SVG elements to be used inside a document. For example, a very simple document using some of the minimal syntax features could look like:

- `<!doctype html>`
- `<title>SVG in text/html</title>`
- `<p>`
- A red circle:
- `<svg> <circle r="50" cx="50" cy="50" fill="red"/> </svg>`
- `</p>`

New semantics

HTML 5 recognises that web pages have a structure, just like books have a structure. In general, web pages have navigation, body content, and sidebar content plus headers, footers, and other features. And HTML 5 has created tags to support those elements of the page.